

NOT SURE IF SOURCE IS

Sources
&
their
Credibility

CREDIBLE OR NOT

THE PLAN!

- ❑ Research Argument
- ❑ Building a Complex Argument out of Simple Ones
 - ❑ In-Class Exercises (2 & 3 from last week's worksheet)
- ❑ The Credibility of Sources
 - ❑ Addition: Personal Experience

RESEARCH ARGUMENT

- In a **research argument**,
 - you make a **claim** or **assertion** (which can be a single sentence or more) that demands support:
 - E.g., *Climate Change is threatening coastal cities; Toni Morrison's most important novel is Beloved.*
 - Your main claim is the assertion that your whole research argument supports. Some call *this* assertion your thesis.

RESEARCH ARGUMENT

- In a **research argument**,
 - you make a **claim** or **assertion**,
 - you back it with **reasons**
 - A **reason** is an **assertion** that supports a claim:
 - E.g., *[Because] climate change is causing ocean levels to rise; [Because] In Beloved, Morrison's major themes find their fullest development.*

RESEARCH ARGUMENT

- In a **research argument**,
 - you make a **claim** or **assertion**,
 - you back it with **reasons** supported by **evidence**
 - **Evidence** is: data deployed to support a **reason**.
 - Unlike a **claim** or a **reason**, **evidence** is not always framed as an **assertion**:
 - E.g., a data table documenting rising ocean levels over the past decade or quotations from *Beloved* illustrating Morrison's major themes would be forms of evidence.

RESEARCH ARGUMENT

- In a **research argument**,
 - you make a **claim** or **assertion**,
 - you back it with **reasons** supported by **evidence**,
 - acknowledge and respond to other views, and sometimes explain your principles of reason.

- Connecting claims and reasons with **warrants** ("the bridge"; "the connection"):
 - The **warrant** is the general principle connecting claims and reasons.
- *Example:*
 - Even when your readers agree that a reason is true, they may still object that it's not relevant to your claim:
 - "We are facing significantly higher health care costs in Europe and North America (**claim**) because climate change is moving the line of extended hard freezes steadily northward (**reason**).
 - Readers might accept the truth of that reason but question its relevance to the claim, asking: *What do higher health costs have to do with hard freezes?* I don't see the **connection**!

RESEARCH ARGUMENT

1. **Claim:** What do you want me to believe? What's your point?
2. **Reasons:** Why do you say that? Why should I agree?
3. **Warrants:** General principle connecting claims and reasons. How does that follow? What's your logic? Can you explain your reasoning?
4. **Evidence:** How do you know? Can you back it up?
5. Acknowledgment and response.

To answer, you *must offer* a **general principle** that justifies relating your *particular* reason to your *particular* claim:

“When an area has fewer hard freezes, it must pay more to combat new diseases carried by subtropical insects no longer killed by those freezes.”

Like all warrants, this one says that if a general circumstance exists (an area has fewer hard freezes), then we can infer a general consequence (that area will have higher costs to combat new diseases).

When climate change is moving the line of extended hard freezes steadily toward (reason).

- Readers might accept the truth of that reason but question its relevance to the claim, asking: *What do higher health costs have to do with hard freezes? I don't see the connection!*

Like all warrants, this one says that if a general circumstance exists (an area has fewer hard freezes), then we can infer a general consequence (that area will have higher costs to combat new diseases).

The logic behind all warrants is that if a generalization is true or reasonable, then so must be specific instances of it.

When climate change is moving the line of extended hard freezes steadily toward (reason).

- Readers might accept the truth of that reason but question its relevance to the claim, asking: *What do higher health costs have to do with hard freezes?* I don't see the connection!

IN-CLASS EXERCISES



- Read the passage "Research Arguments"
- 2. Analyze the arguments on your worksheet (from last class)

IN-CLASS EXERCISES

1. Regular TV viewers overestimate both the rate of crime in their neighborhood and the personal danger to themselves and their families.
 - Even with its relatively precise language, this claim offers only a single proposition.

IN-CLASS EXERCISES

2. Although violent crime is actually decreasing, regular TV viewers overestimate their neighborhood crime rate by 150 percent and therefore misjudge personal danger to themselves and their families because local TV evening news regularly opens with graphic reports of mayhem and murder in familiar locations, making many believe that crime happens nightly outside their front door.

- It makes the logic explicit. While that claim may seem overwritten, it foreshadows three of the five elements that you need for a full argument: **(1) Although I acknowledge X, (2) I claim Y (3) because of reason Z.**

IN-CLASS EXERCISES

3. Elementary schools should make teaching foreign languages a priority because we acquire languages best and most easily when we are young.

- Elementary schools should make teaching foreign languages a priority (claim) because we acquire languages best and most easily when we are young (reason).

IN-CLASS EXERCISES

4. Elementary schools should make teaching foreign languages a priority because we acquire languages best and most easily when we are young. In fact, those who begin second languages as adults rarely attain the level of fluency of those who learn them as children. Teaching foreign languages at the elementary-school level also contributes to children's ethical development, because it fosters an awareness of cultures and societies beyond their own.

- (Reasons may require further support in complex arguments). Elementary schools should make teaching foreign languages a priority (**claim 1**) because we acquire languages best and most easily when we are young (reason 1 supporting **claim 1 / claim 2**). In fact, those who begin second languages as adults rarely attain the level of fluency of those who learn them as children (reason 2 supporting reason 1 and **claim 2 / claim 3**). Teaching foreign languages at the elementary-school level also contributes to children's ethical development (reason 3 supporting **claim 1 / claim 4**), because it fosters an awareness of cultures and societies beyond their own (reason 4 supporting reason 3 and **claim 4 / claim 5**).

5. Elementary schools should make teaching foreign languages a priority because we acquire languages best and most easily when we are young. In fact, those who begin second languages as adults rarely attain the level of fluency of those who learn them as children. In a study of over one hundred second-language learners, Jones (2013) identified an inverse correlation between second-language proficiency and ...(see table 1).

- 5. Elementary schools should make teaching foreign languages a priority (**claim 1**) because we acquire languages best and most easily when we are young (reason 1 supporting **claim 1** / **claim 2**). In fact, those who begin second languages as adults rarely attain the level of fluency of those who learn them as children (reason 2 supporting **reason 1** / **claim 3**). In a study of over one hundred second-language learners, Jones (2013) identified an inverse correlation between second-language proficiency and ...(see table 1) (evidence supporting **reason 2**).

6. "...despite their selfish belligerence, humans' rationality manifests itself in their capacity to understand the benefits of peace and order and, therefore, in their willingness to agree to set up an association, a State, a government which will facilitate peaceful coexistence. However, such an agreement does not suffice ... Even though people recognise that peace is better than war and universal respect of neighbours is preferable to feuding, each individual also recognises that his or her desires are best served by being unprincipled in a principled community. And if everyone prefers that all others are law-abiding while he or she cheats, steals, coerces and murders, then the agreement between people will be worthless and the 'war of all against all' will recommence. What is needed ...is something stronger than some agreement; individuals must agree to give up some of their rights to a higher authority (e.g. the State) so as to live in peace. " (Yanis Varoufakis, Foundations of Economics)

- "...despite their selfish belligerence, humans' rationality manifests itself in their capacity to understand the benefits of peace and order and, therefore, in their willingness to agree to set up an association, a State, a government which will facilitate peaceful coexistence (**claim 1**). However, such an agreement does not suffice (**claim 2**)... Even though people recognise that peace is better than war and universal respect of neighbours is preferable to feuding, each individual also recognises that his or her desires are best served by being unprincipled in a principled community (**claim 3**). And if everyone prefers that all others are law-abiding while he or she cheats, steals, coerces and murders, then the agreement between people will be worthless and the 'war of all against all' will recommence (reason 1 supporting **claims 4 and 2 / claim 4**). What is needed ...is something stronger than some agreement (restatement of **claim 2**); individuals must agree to give up some of their rights to a higher authority (e.g. the State) so as to live in peace. (reason 2 supporting **claim 2 / claim 5**)."

7. American higher education must curb escalating tuition costs, because the price of college is becoming an impediment to realizing the American dream. Today a majority of students leave college with a crushing debt burden. In 2013, nearly 70 percent of students borrowed money for college with loans averaging \$30, 000, a debt that prevents many from buying a home, beginning a family, or pursuing a higher degree.

- American higher education must curb escalating tuition costs (claim 1), because the price of college is becoming an impediment to realizing the American dream (reason). Today a majority of students leave college with a crushing debt burden (soft evidence). In 2013, nearly 70 percent of students borrowed money for college with loans averaging \$30, 000, a debt that prevents many from buying a home, beginning a family, or pursuing a higher degree (hard evidence).
 - Note, a real sceptical reader could ask again: What backs up those numbers? What justifies the claim that this situation is a crisis?
 - If so, the reader would have to provide still harder data, breaking down those numbers to document the consequences of debt for recent graduates. If she did her own research, she could show her raw data. If she found her data in a secondary source, she could cite it, but she might then be asked to prove that her source is reliable. Really sceptical readers just never give up.

8. When an area has fewer hard freezes, it can expect higher medical costs to cope with diseases carried by subtropical insects that do not survive freezes. Europe and North America must thus expect higher health care costs because climate change is moving the line of extended hard freezes steadily north. In the last one hundred years, the line of hard freezes lasting more than two weeks has moved north at the rate of roughly...

- When an area has fewer hard freezes, it can expect higher medical costs to cope with diseases carried by subtropical insects that do not survive freezes. (*general principle a.k.a. warrant*) Europe and North America must thus expect higher health care costs (main claim) because climate change is moving the line of extended hard freezes steadily north (reason). In the last one hundred years, the line of hard freezes lasting more than two weeks has moved north at the rate of roughly... (evidence)

9. A whale is more closely related to a hippopotamus than to a cow, because it shares more DNA with a hippopotamus. When a species shares more DNA with one species than it does with another, we infer that it is more closely related to the first.

- **(Provide warrants).** A whale is more closely related to a hippopotamus than to a cow **(claim)**, because it shares more DNA with a hippopotamus **(reason)**. When a species shares more DNA with one species than it does with another (circumstances), we infer that it is more closely related to the first (consequence).
- Last sentence is the warrant: “whether or not a warrant gets stated explicitly depends not only on the argument but also on the audience. Expert states principles that are obvious to other experts only when they communicate with non-experts – or when challenged.

10. The Russian Federation faces a falling standard of living, because its birthrate is only 13.2 per 1, 000 and life expectancy for men is only about 63 years.

- **Missing warrants.** The Russian Federation faces a falling standard of living (claim), because its birthrate is only 13.2 per 1, 000 and life expectancy for men is only about 63 years (reason).
- Note: A connection is *needed* between claim and reason.

11. The Russian Federation faces a falling standard of living, because its birthrate is only 13.2 per 1, 000 and life expectancy for men is only about 63 years. When a nation's labor force shrinks, its economic future is grim.

- The Russian Federation faces a falling standard of living (**claim**), because its birthrate is only 13.2 per 1, 000 and life expectancy for men is only about 63 years (**reason**). When a nation's labor force shrinks (**general circumstances**), its economic future is grim (**general consequence**).
- Last sentence is the warrant.

12. Contrary to popular belief, gun ownership in America was probably not widespread in the first half of the nineteenth century and before, because guns were rarely mentioned in wills. A review of 4, 465 wills filed in seven states from 1750 to 1850 shows that only 11 percent mention a long gun or handgun.

- Contrary to popular belief, gun ownership in America was probably not widespread in the first half of the nineteenth century and before (claim), because guns were rarely mentioned in wills (reason). A review of 4, 465 wills filed in seven states from 1750 to 1850 shows that only 11 percent mention a long gun or handgun (report of evidence).
- Note: Such as claim is likely to be resisted by those who believe that those who founded the United States owned guns. So even if they accept that the reason is true – that guns were in fact rarely mentioned in wills – they may still object: *But I don't see how that counts as a reason to believe that few people owned a gun. It's irrelevant.*

13. In the eighteenth and early nineteenth centuries, valuable objects were listed in wills, so when someone failed to mention a valuable object in his will, he did not own one. Since guns were valuable but were rarely mentioned in wills before 1850, gun ownership must not have been widespread.

- In the eighteenth and early nineteenth centuries, valuable objects were listed in wills, so when someone failed to mention a valuable object in his will, he did not own one (*warrant*). Since guns were valuable but were rarely mentioned in wills before 1850 (*reason*), gun ownership must not have been widespread (*claim*).

14. In the eighteenth and early nineteenth centuries, valuable objects were listed in wills, so when a will failed to mention such a valuable object, the person did not own one. Watson (1989) confirmed that to be the case. In a study of 1,356 wills filed in Cumberland County between 1750 and 1825, he found ...

- **Note:** *A warrant seems reasonable when readers can accept that its consequence follows from its circumstances. If readers don't accept that, you must first **convince them to accept your warrant by treating it as a claim in its own argument**, supported by its own reason and evidence.*
 - In the eighteenth and early nineteenth centuries, valuable objects were listed in wills, so when a will failed to mention such a valuable object, the person did not own one (**warrant/claim**). Watson (1989) confirmed that to be the case (**reason**). In a study of 1,356 wills filed in Cumberland County between 1750 and 1825, he found ...(**evidence**)
 - Note: "In the eighteenth and early nineteenth centuries, valuable objects were listed in wills." That version is too broad; it might seem more plausible if it were qualified: "In the eighteenth and early nineteenth centuries, most household objects considered valuable by their owners were usually listed in wills."

15. Children aged 12-16 today are significantly more violent than their counterparts from a generation ago. Brown (2013) has shown that... Given these facts, it seems highly likely that violent video games are exerting a destructive influence on today's youth.

- All arguments rely on warrants, even if they aren't stated explicitly. You can test the **soundness** of an argument by trying to imagine a warrant for it. Here is a **flawed argument** about how violent games affect children.
 - Children aged 12-16 today are significantly more violent than their counterparts from a generation ago (reason). Brown (2013) has shown that... (evidence) Given these facts, it seems highly likely that violent video games are exerting a destructive influence on today's youth (claim).

16. When children are constantly exposed to images of sadistic violence, they are influenced for the worse. Children aged 12–16 today are significantly more violent than their counterparts from a generation ago. Brown (2013) has shown that... Given these facts, it seems highly likely that violent video games are exerting a destructive influence on today's youth.

- Note: To understand what's wrong here, we can imagine a warrant that would allow us to draw a conclusion about the effect of video games on children.
- When children are constantly exposed to images of sadistic violence, they are influenced for the worse (**warrant**). Children aged 12–16 today are significantly more violent than their counterparts from a generation ago (**reason**). Brown (2013) has shown that...(evidence) Given these facts, it seems highly likely that violent video games are exerting a destructive influence on today's youth (**claim**).
- Now we can see the problem: the specific circumstance – rising violence among children aged 12-16 – is not a valid instance of the warrant's general circumstance: children being exposed to images of sadistic violence.

17. When children are constantly exposed to images of sadistic violence, they are influenced for the worse. Over the past decade, video games have become a major source of children's exposure to violent imagery. Jones (2014) shows that... Given these facts, it seems highly likely that violent video games are exerting a destructive influence on today's youth, making them significantly more violent than their counterparts from a generation ago.

- Note: So even if that statement about rising violence among children is true, it can't justify the claim. To fix that argument, we have to **revise** the **reason** to be a good instance of the warrant's general circumstance, which may also mean producing **new evidence** to support that **new reason**.
 - When children are constantly exposed to images of sadistic violence, they are influenced for the worse (**warrant**). Over the past decade, video games have become a major source of children's exposure to violent imagery (**new reason**). Jones (2014) shows that... (**new evidence**) Given these facts, it seems highly likely that violent video games are exerting a destructive influence on today's youth, making them significantly more violent than their counterparts from a generation ago (**claim**).
- Now the reason and claim seem closer to what the warrant covers or includes. But a reader keen to derail the argument might still object: *Wait. All those images aren't sadistic! Much of it is cartoon violence. And children aren't constantly exposed to it.*

BUILDING A COMPLEX ARGUMENT OUT OF SIMPLE ONES



1. **Claim:** What do you want me to believe? What's your point?
2. **Reasons:** Why do you say that? Why should I agree?
3. **Warrants:** General principle connecting claims and reasons. How does that follow? What's your logic? Can you explain your reasoning?
4. **Evidence:** How do you know? Can you back it up?
5. Acknowledgment and response.

BUILDING A COMPLEX ARGUMENT OUT OF SIMPLE ONES

- These five elements constitute a “basic” argument, but research arguments are more complex:
 - We almost always *support a claim with two or more reasons, each of which must be supported by its own additional reasons and evidence* and perhaps *justified by its own warrant*. Since readers think of many alternatives and objections to any complex argument, careful researchers typically have to respond to a number of them.
 - Moreover, *each reason, warrant, or response to an objection* (all of which are statements or assertions) *may itself have to be treated as subclaim and supported by its own argument*. Only the evidence “stands alone,” but even then you may have to explain where you got it, why you think it’s reliable, and how it supports your reasons- and that may require another argument.
 - And finally, most arguments also include background, definitions, explanations of issues, etc.

BUILDING A COMPLEX ARGUMENT OUT OF SIMPLE ONES

- One more thing: You make your argument stronger and more credible by modestly acknowledging its limits.
- Limit your claims to what your argument can actually support by qualifying their scope and certainty:

IN-CLASS EXERCISES

- 3. Analyze the arguments on your worksheet (from last class)

IN-CLASS EXERCISES

1. Elementary schools should make foreign languages a priority because we acquire languages best and most easily when we are young. ... Of course, if schools increase the attention they give to foreign languages, quality of instruction in other subjects might decline. But little evidence exists to support that fear and much dispels it...
 - Elementary schools should make foreign languages a priority (**claim 1**) because we acquire languages best and most easily when we are young. (**reason 1 supporting claim 1 / claim 2**) ... Of course, if schools increase the attention they give to foreign languages, quality of instruction in other subjects might decline (**acknowledgment**). But little evidence exists to support that fear and much dispels it... (**response**)

IN-CLASS EXERCISES

2. Some have argued that food can be addictive, but we are concerned here only with substances for which addiction is the norm.

- Some have argued that food can be addictive (acknowledgment of objection), but we are concerned here only with substances for which addiction is the norm (reason why objection is irrelevant).
- Note: That initial response may be enough, but only if readers recognize the basis for it, either because it's obvious or because you've made the argument before. Otherwise, explain its basis using additional reasons and evidence.

IN-CLASS EXERCISES

3. Some have argued that food can be addictive, but we are concerned here only with substances for which addiction is the norm. Some who taste chocolate once may be unable to resist it thereafter, but their number is a fraction of those who are immediately addicted to crack cocaine after a single exposure. Chernowitz (1998) found that just one exposure resulted in ...

- Some have argued that food can be addictive (acknowledgment of objection), but we are concerned here only with substances for which addiction is the norm (reason why the objection is irrelevant). Some who taste chocolate once may be unable to resist it thereafter, but their number is a fraction of those who are immediately addicted to crack cocaine after a single exposure (subreason). Chernowitz 1998) found that just one exposure resulted in ...(report of evidence)

IN-CLASS EXERCISES

4. Although many believe that school uniforms help lower the incidence of violence in public schools, the evidence is at best weak because researchers have not controlled for other measures that have been instituted at the same time as uniforms and because the data reported are statistically suspect.

- Although many believe that school uniforms help lower the incidence of violence in public schools (qualifications), the evidence is at best weak (**claim**) because researchers have not controlled for other measures that have been instituted at the same time as uniforms (**reason 1**) and because the data reported are statistically suspect (**reason 2**).
- **Note:** *If readers might think of those qualifications, acknowledge them first. You not only imply that you understand their views, but commit yourself to responding to them in the course of your argument.*

5. Today Franklin D. Roosevelt is revered as one of our most admired historical figures, but toward the end of his second term, he was quite unpopular , at least among certain segments of American society. Newspapers, for example, attacked him for promoting socialism, a sign that a modern administration is in trouble. In 1938, 70 percent of Midwest newspapers accused him of wanting the government to manage the banking system...Some have argued otherwise, including Nicholson (1983, 1992) and Wiggins (1973), both of whom offer anecdotal reports that Roosevelt was always in high regard, but these reports are supported only by the memories of those who had an interest in deifying FDR. Unless it can be shown that the newspapers critical of Roosevelt were controlled by special interests, their attacks demonstrate significant popular dissatisfaction with Roosevelt's presidency.

- Today Franklin D. Roosevelt is revered as one of our most admired historical figures, but toward the end of his second term, he was quite unpopular (**main claim**), at least among certain segments of American society. Newspapers, for example, attacked him for promoting socialism, a sign that a modern administration is in trouble (**limitation of claim**). In 1938, 70 percent of Midwest newspapers accused him of wanting the government to manage the banking system...Some have argued otherwise, including Nicholson (1983, 1992) and Wiggins (1973), both of whom offer anecdotal reports that Roosevelt was always in high regard (acknowledgment to objection), but these reports are supported only by the memories of those who had an interest in deifying FDR (response). Unless it can be shown that the newspapers critical of Roosevelt were controlled by special interests (**limitation of claim**), their attacks demonstrate significant popular dissatisfaction with Roosevelt's presidency (**restatement of claim**).



Take a deep breath!

NOT SURE IF SOURCE IS

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THE CREDIBILITY OF SOURCES



- What kinds of sources of information are available when researching a scientific claim? How can we use them to gather as much useful information as possible

Direct Observation



Good Arguments

- 1. If it's raining outside, then it's cloudy. (True)
- 2. It's not cloudy. (True)
- Conclusion: Therefore, it's not raining. (True)

Reliable Sources



THE CREDIBILITY OF SOURCES

- We turn not from the credibility of claims themselves to the credibility of the sources from which we get them.
- Terminology:
 - Credence = degree of belief
 - How plausible you find a claim.
- We are automatically suspicious of certain sources of information.
- We'll look at several factors that should influence how much credence we give to a source.

THE CREDIBILITY OF SOURCES



Reflect & Discuss:

- What factors should one consider when assessing the relevance of a source?

THE CREDIBILITY OF SOURCES

- We start with an important rule for deciding whom to trust.
- Our rule makes use of two correlative concepts, *interested* and *disinterested parties*:
 - A person who stands to gain from our belief in a claim is known as an *interested party*, and *interested parties* must be viewed with much more suspicion than *disinterested parties*, who have no stake in our belief one way or another.
 - *Can you think of an example of an interested party?*

THE CREDIBILITY OF SOURCES

- Furthermore, we often base our judgments on irrelevant considerations.
 - *Physical* characteristics, for instance, tell us little about a person's credibility or its lack.
 - Does the person look you in the eye? Does the person have a nervous laugh? ... Despite being generally worthless in this regard, such characteristics are widely used in sizing up a person's credibility (e.g., con artist).

THE CREDIBILITY OF SOURCES

- Furthermore, we often base our judgments on irrelevant considerations.
 - *Physical* characteristics, for instance, tell us little about a person's credibility or its lack.
 - Other *irrelevant* characteristics we sometimes use to judge a person's credibility include gender, age, ethnicity, accent, mannerisms...

THE CREDIBILITY OF SOURCES



- Read the passage “Critical Discourse,” making note of any important points and definition
- **Reflect & Discuss:**
 - Describe a few projects which can be classified as ‘big science’.
 - Invent a scientific issue, and then identify someone who would likely be an interested party regarding that issue; then identify someone who is likely not to be an interested party.

THE CREDIBILITY OF SOURCES



Reflect & Discuss:

- What factors should one consider when assessing the reliability of a source?

THE CREDIBILITY OF SOURCES

Expertise:

- Two indicators considered minimal prerequisites for being considered an expert:
 - Education and training from reputable institutions or programs in the relevant field (evidenced by degrees, etc).
 - ***Experience*** in making ***reliable judgments*** in the field (the more experience the better).

THE CREDIBILITY OF SOURCES

Expertise:

- Two indicators considered minimal prerequisites for being considered an expert:
 - Education and training from reputable institutions or programs in the relevant field (evidenced by degrees, etc).
 - *Experience* in making *reliable judgments* in the field (the more experience the better).
 - *Reputation* among peers.
 - Professional *accomplishments*.

THE CREDIBILITY OF SOURCES

Expertise:

- Note:
 - *Just because someone is an expert in one field doesn't mean he or she is an expert in another field.*
- Important:
 - it's not that they can't be right,
 - it's that ***expertise in one field*** should not be reason to believe what is said about an ***unrelated*** field

THE CREDIBILITY OF SOURCES

- Important to state:
- *We are often justified* in believing an unsupported claim because it is based on expert opinion.
 - But: this is not because of the strength of direct reasons that we have
 - *It is based on the reputation of that expert or group of experts*

THE CREDIBILITY OF SOURCES

- When should we doubt expert opinion?
 - Bias: motivations for something other than truth (financial gain, loyalty to a cause, professional ambition, emotional needs, political outlook, personal ideology, etc).
- *So: if we have good reason to believe an expert is biased, we are not justified in believing their opinion without further investigation.*

THE CREDIBILITY OF SOURCES

- Tell-tale signs?
 - 1. Expert is being paid or funded by a special interest group or company
 - 2. When an expert expresses strong belief (or disbelief) even though there is no evidence to support their belief
 - 3. When the expert stands to gain financially.

THE CREDIBILITY OF SOURCES

- Approach these with a critical outlook:
 - Has the expert been reliable in the past?
 - What do his/her peers think?
 - Are their results always in concert with COIs?

THE CREDIBILITY OF SOURCES



- Read the passage “Social Science”, making notes of any important points and definitions.
- Anything interesting you would like to share?

ADDITION: PERSONAL EXPERIENCE

THE CREDIBILITY OF SOURCES

- Addition: *Personal Experience*
- We accept many claims because they are based on *personal experience*—our own *or* someone else's.
 - Personal experience = arising from senses, memory, and judgment

THE CREDIBILITY OF SOURCES

- You believe that Jack caused the traffic accident because you, or someone else, witnessed it.
 - You think that the herbal tea cured your headache because the pain went away after you drank it.
 - You believe that your friend can bend spoons with her mind because you saw her do it at a party.
 - As a member of a jury, you vote to convict the defendant because eyewitness testimony puts him at the scene of the crime with a gun in his hand.
- But can you trust personal experience to reveal the truth?

THE CREDIBILITY OF SOURCES

- A principle:
 - *It's reasonable to accept the evidence provided by personal experience only if there's no good reason to doubt it.*

THE CREDIBILITY OF SOURCES

- A principle:
 - *It's reasonable to accept the evidence provided by personal experience only if there's no good reason to doubt it.*
- If our faculties *are working properly* and *our use of them is unimpeded* by anything in our environment, we're entitled to accept what our personal experience tells us.

THE CREDIBILITY OF SOURCES

- Of course, our personal experience is *not infallible*.
- Under certain circumstances, our senses, memory, and judgment can't be trusted.

THE CREDIBILITY OF SOURCES

- The following factors can give us good reason to doubt the reliability of personal experience:
 - 1. Impairment
 - 2. Expectation
 - 3. Innumeracy
- And the same goes for *testimony*

THE CREDIBILITY OF SOURCES

- **Impairment**
- If our perceptual powers are somehow impaired or impeded, we have reason to doubt them.
 - For example: when our senses are debilitated: we are ill, tired, stressed, drunk, disoriented, distracted, etc
 - Or when we are in a situation that interferes with sensory input: when our environment is too dark, too light, too noisy, too hazy, etc

THE CREDIBILITY OF SOURCES

- Impairment
- And *memory* can be affected by many of the same factors
 - Memories can be distorted if formed during times of stress
 - This can complicate eye witness testimony for horrible crimes

THE CREDIBILITY OF SOURCES

- Impairment
- Important:
 - Our perception and memory are *not like recording devices*: they do not make exact mental copies
 - They are *constructive*: they use bits of sensory input to create representations of things, not replicas
 - What we perceive and remember is to some degree fabricated by our minds.

THE CREDIBILITY OF SOURCES

- Impairment
- Example:

Once upon a time, a man (whom we'll call Mike) stumbled upon an armed robbery in a hardware store. The robber rummaged around the cluttered store brandishing a silver weapon; finally, he stole all the money. Then, almost as an afterthought, he grabbed a hand calculator and a hammer, putting these in his satchel as he left the store. The police were summoned immediately, but before they arrived Mike talked to another customer about the robbery. We'll call her Maria. Maria told Mike that she saw the robber grab a calculator and a screwdriver, stuffing them in his satchel as he left the store. The police arrived, and when they questioned Mike he recounted the robbery at some length. He described in detail the silver weapon, the money, and the calculator. When the police asked him about a tool that they heard had been taken, "Did you see if it was a hammer or a screwdriver?" he said, "Screwdriver."³

THE CREDIBILITY OF SOURCES

- Expectation
- Sometimes we perceive exactly what we expect to perceive—regardless of whether there's anything there to detect.

THE CREDIBILITY OF SOURCES

- **Expectation**
- Research indicates that expectation has a more powerful effect on our experience than most think.
 - In lots of cases, the *mere suggestion* that the stimuli would occur was enough to cause subjects to report perceiving things that did not in fact exist.
 - This is called the “*expectancy effect*” and it has many applications

THE CREDIBILITY OF SOURCES

- Expectation
- <https://www.youtube.com/watch?v=feBrQ5A-UUE>

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- **Expectation**
- This is especially pronounced when the stimuli are *vague* or *ambiguous*.
- "Pareidolia"
 - A kind of illusion: the tendency to interpret a vague stimulus as something known to the observer



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- **Expectation**
 - Strong expectancy effects indicate the need to double check our sensory information, pause in our haste to draw conclusions.

THE CREDIBILITY OF SOURCES

- Innumeracy
 - We are bad at using numbers and math
 - When we make an off-the-cuff judgment about the chances of something happening, we should be extra careful because we humans are terrible at estimating probabilities.

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Here's a classic example. Imagine there are 23 students, including you, in your classroom. What are the chances that at least 2 of the students have exactly the same birthday? (Not the same date of birth, but the same birthday out of the 365 possible ones.) Most people are surprised to find out that the answer is neither 1 chance in 365 ($1/365$), nor 1 in 52 ($1/52$). It's 1 chance in 2 ($1/2$, or 50–50)—a completely counterintuitive result. Math can give you the right answer, but guessing or estimating is very unlikely to do so.

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- Innumeracy
 - Another common error is *misjudging coincidences as the belief* that something else (paranormal or supernatural) must be going on...

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event may be highly probable over the long haul. It may be unlikely in any given instance for you to flip a coin and get tails seven times in a row. But this “streak” is virtually certain to happen if you flip the coin enough times.

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- Innumeracy
- Another error is called the *gambler's fallacy*.
 - The error of thinking that previous events can affect the probabilities in the random event at hand.

On the first toss, the odds are, of course, one in two, or 50–50, that it will land tails. It lands tails. Astoundingly, on the other five tosses the coin also lands tails. That's six tails in a row. So what are the odds that the coin will land tails on the seventh toss? Answer: 50–50. Each toss has exactly the same probability of landing tails (or heads): 50–50. The coin does not remember previous tosses, and to think otherwise is to commit the gambler's fallacy. You see it a lot in casinos, sporting events, and—alas—everyday decision making.

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- Innumeracy
 - In other words, we're pretty bad with numbers when we're reasoning intuitively

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REVIEW NOTES

Personal Experience

- It's reasonable to accept the evidence provided by personal experience only if there's no good reason to doubt it.
- If our perceptual powers are impaired or impeded, we have reason to doubt them.
- Our perception and memory are constructive, which means that our minds are capable of manufacturing what we seem to have experienced.
- We often perceive exactly what we expect to perceive, and this tendency is enhanced when stimuli are vague or ambiguous.
- The gambler's fallacy is the mistake of thinking that previous events can affect the probabilities in the random event at hand.